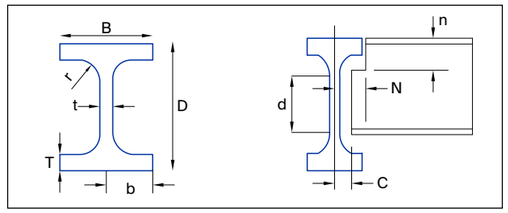
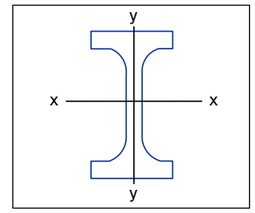
2.3
Universal Beams & Columns (UB & UC)



Imperial Units

Designation			Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre				Flange	Web				Flange	Web	End clearance	Notch	
			D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W12-12x4 (305x102)	14	20.8	302.5	100.8	5.7	5.1	7.6	275.8	26.8	8.83	54.3	5	58	13
	16	23.8	304.5	101.3	6.7	5.6	7.6	275.8	30.4	7.53	49.3	5	58	14
	16.5	24.8	305.1	101.6	7.0	5.8	7.6	275.9	31.6	7.26	47.6	5	58	15
	19	28.2	308.7	101.8	8.8	6.0	7.6	275.9	35.9	5.78	46.0	5	58	16
	22	32.8	312.7	102.4	10.8	6.6	7.6	275.9	41.8	4.74	41.8	5	58	18
W12-12x5 (305x127)	25	37.0	304.4	123.3	10.7	7.1	8.9	265.2	47.2	5.76	37.4	6	68	20
	28	41.9	307.2	124.3	12.1	8.0	8.9	265.2	53.4	5.14	33.2	6	68	21
	32	48.1	311.0	125.3	14.0	9.0	8.9	265.2	61.2	4.48	29.5	7	68	23
W12-12x6 1/2 (305x165)	26	38.7	310.4	164.8	9.7	5.8	7.6	275.9	49.3	8.54	47.2	5	89	17
	27	40.3	303.4	165.0	10.2	6.0	8.9	265.2	51.3	8.09	44.2	5	90	19
	30	44.6	313.4	165.6	11.1	6.6	7.6	276.0	56.5	7.46	41.8	5	90	19
	31	46.1	306.6	165.7	11.8	6.7	8.9	265.2	58.7	7.02	39.6	5	90	21
	35	52.1	317.5	166.6	13.2	7.6	7.6	275.9	66.7	6.31	36.2	6	89	21
	36	54.0	310.4	166.9	13.7	7.9	8.9	265.2	68.8	6.09	33.6	6	90	23
W12-12x8 (305x203)	40	59.5	303.3	203.3	13.0	7.5	15.2	246.9	75.6	7.82	33.0	6	108	28
	45	67.0	306.3	204.3	14.6	8.5	15.2	246.7	85.2	7.00	29.0	6	108	30
	50	74.4	309.6	205.2	16.2	9.4	15.2	246.8	94.5	6.33	26.3	7	108	31
W12-12x10 (305x254)	53	78.9	306.3	253.9	14.6	8.8	15.2	246.7	100	8.70	28.2	6	133	30
	58	86.3	309.6	254.3	16.2	9.1	15.2	246.8	110	7.85	27.0	7	133	31
W12-12x12 (305x305)	65	96.9	307.9	305.3	15.4	9.9	15.2	246.7	123	9.91	24.9	7	158	31
	72	107.1	311.1	305.8	17.0	10.9	15.2	246.7	136	8.98	22.6	7	157	32
	79	117.9	314.5	307.4	18.7	12.0	15.2	246.7	150	8.22	20.6	8	158	34
	87	129.5	318.3	308.0	20.6	13.1	15.2	246.8	165	7.49	18.9	9	157	36
	92	136.9	320.5	309.2	21.7	13.8	15.2	246.7	174	7.12	17.9	9	158	37
	96	142.8	322.8	308.9	22.9	14.0	15.2	246.7	182	6.76	17.7	9	157	38
	106	158.1	327.1	311.2	25.0	15.8	15.2	246.7	201	6.22	15.6	10	158	40
	120	178.6	333.2	312.9	28.1	18.0	15.2	246.7	228	5.57	13.7	11	157	43
	133	198.1	339.9	314.5	31.4	19.1	15.2	246.7	252	5.01	12.9	12	158	47
	136	202.4	340.6	315.0	31.8	20.1	15.2	246.7	258	4.96	12.3	12	157	47
	152	226.2	348.2	317.0	35.6	22.1	15.2	246.7	289	4.46	11.2	13	157	51
	161	240	352.5	318.4	37.7	23.0	15.2	246.7	306	4.22	10.7	14	158	53
	170	253	356.4	319.3	39.6	24.4	15.2	246.8	323	4.03	10.1	14	157	55

Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)



Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
0.985	3682	97.9	11.7	1.91	243	19.4	286	31.1	0.842	50.8	0.022	2.93
0.990	4274	117	11.9	1.96	281	23.1	329	37.0	0.846	45.0	0.026	4.27
0.992	4455	123	11.9	1.97	292	24.2	342	38.8	0.846	43.4	0.027	4.77
1.00	5366	155	12.2	2.08	348	30.5	403	48.5	0.859	37.4	0.035	7.40
1.01	6501	194	12.5	2.15	416	37.9	481	60.0	0.866	31.6	0.044	12.2
1.07	7166	335	12.3	2.67	471	54.4	539	85.3	0.872	29.7	0.072	14.8
1.08	8196	389	12.4	2.70	534	62.6	614	98.4	0.872	26.5	0.085	21.1
1.09	9575	461	12.5	2.74	616	73.6	711	116	0.873	23.3	0.102	31.8
1.26	8498	720	13.1	3.82	548	87.4	609	134	0.887	33.8	0.163	12.5
1.24	8503	764	12.9	3.86	560	92.7	623	142	0.889	31.0	0.164	14.7
1.26	9864	841	13.2	3.86	629	102	703	156	0.887	29.7	0.192	18.7
1.25	9899	896	13.0	3.90	646	108	720	166	0.891	27.1	0.195	22.2
1.27	11860	1019	13.3	3.91	747	122	838	188	0.889	25.3	0.236	30.8
1.26	11700	1063	13.0	3.93	754	127	846	196	0.889	23.6	0.234	34.8
1.38	12840	1823	13.0	4.91	847	179	938	274	0.890	22.9	0.384	39.1
1.39	14570	2078	13.1	4.94	952	203	1060	311	0.889	20.6	0.442	54.6
1.40	16350	2337	13.2	4.97	1056	228	1183	349	0.889	18.8	0.503	73.5
1.58	17700	3986	13.3	6.30	1156	314	1276	477	0.876	20.4	0.848	65.6
1.59	19740	4444	13.4	6.36	1275	349	1411	531	0.878	18.7	0.956	86.5
1.79	22250	7308	13.4	7.69	1445	479	1592	726	0.850	19.3	1.56	91.2
1.80	24830	8117	13.5	7.72	1596	531	1767	806	0.850	17.6	1.75	122
1.81	27670	9060	13.6	7.77	1760	589	1958	895	0.850	16.2	1.98	161
1.82	30810	10030	13.7	7.80	1936	651	2164	990	0.851	14.9	2.22	212
1.82	32820	10700	13.7	7.83	2048	692	2297	1053	0.851	14.2	2.39	249
1.83	34670	11240	13.8	7.86	2148	728	2413	1106	0.853	13.6	2.53	285
1.84	38750	12570	13.9	7.90	2369	808	2680	1230	0.851	12.5	2.87	378
1.86	44560	14350	14.0	7.94	2675	917	3053	1399	0.851	11.2	3.34	537
1.87	50900	16300	14.2	8.04	2995	1037	3440	1581	0.853	10.2	3.88	734
1.87	51790	16560	14.2	8.02	3041	1052	3501	1606	0.852	10.1	3.95	771
1.89	59610	18910	14.4	8.09	3424	1193	3975	1823	0.853	9.2	4.62	1074
1.91	64200	20320	14.5	8.15	3643	1276	4247	1951	0.854	8.7	5.03	1271
1.92	68490	21540	14.6	8.17	3844	1349	4502	2064	0.854	8.4	5.40	1480